

The rate of wound healing is increased in psoriasis

Psoriasis shares many features with wound healing, a process that involves switching keratinocytes from growth to differentiation. Calcium is known to regulate this process. The N-methyl-D-aspartate receptor (NMDAR), an ionotropic glutamate receptor found on keratinocytes, is expressed abnormally in psoriasis *in vivo*. Morhenn et al investigated whether the rate of healing in the skin of psoriatic individuals differs from that observed in normal skin and whether the keratinocyte hyperproliferation found in psoriasis correlates with expression of specific NMDAR subunits. On day 0, as well as, on day 6 after the 3-mm biopsy, photographs were taken and the size of the wounds determined using ImageJ. Involved and uninvolved skin of individuals with psoriasis shows significantly more rapid healing than normal. The NR2C subunit of NMDAR is down-regulated in the basal cell layer of involved and uninvolved epidermis of psoriatic subjects compared to controls. By contrast, cells in the basal cell layer of the uninvolved epidermis showed a significantly greater percent strong staining for NR2D.

Thus, wound healing is significantly accelerated in psoriasis compared to normal.

Epithelial to mesenchymal transition markers are associated with an increased metastatic risk in primary cutaneous squamous cell carcinomas but are attenuated in lymph node metastases

It is known that epithelial to mesenchymal transition (EMT) is a process involving loss of intercellular adhesion, acquisition of a mesenchymal phenotype and enhanced migratory potential. Epithelial markers, such as E-cadherin, are downregulated and mesenchymal proteins (Vimentin), increased in association with EMT. The involvement of EMT in cutaneous squamous cell carcinoma (SCC) has been proposed, but remains unclear. Toll et al investigated the expression of EMT markers in metastatic SCC (MSCC) and their corresponding metastases, and correlated them with clinico-pathological factors associated with an increased risk of metastasis by a retrospective study that included 146 cutaneous SCC samples. Loss of membranous E-cadherin was observed in 77% SCCs, with no differences between MSCC and non-MSCC. No significant Snail1 expression was detected. Twist, Zeb1, Vimentin, β -catenin and Podoplanin were significantly overexpressed in MSCCs. Twist ectopic expression in SCC13 cells induced Zeb1, Vimentin and Podoplanin expression and E-cadherin

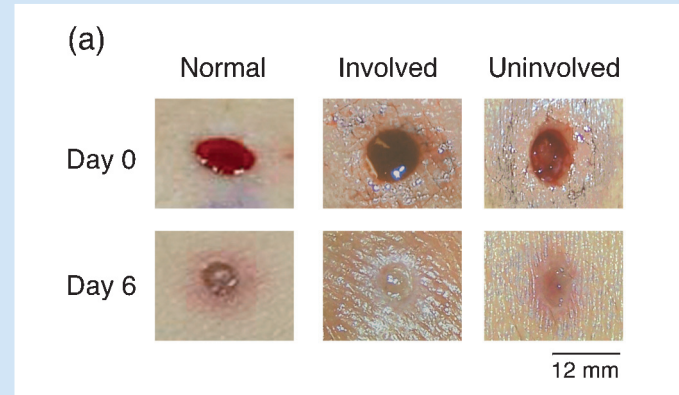


Fig. 1. Measurement of wound healing in psoriasis and normal skin. (a) Photographs of the sites of the biopsies at days 0 and 6 performed in normal, involved and uninvolved skin.

delocalization. Expression of EMT markers was decreased in the metastases when compared with the corresponding primary tumors. Thus, apartial EMT, characterized by the expression of Twist but without a total E-cadherin depletion, is involved in the acquisition of invasive traits by cSCC, but the process is downregulated in lymph node metastases.

Table 2

Univariate analysis of immunohistochemical risk factors for lymphatic metastases

	Non-MSCC (N = 51)	MSCC (N = 56)	p-Value
Membranous E-cadherin underexpression	36/49 (73.5%)	45/56 (80.4%)	0.487
Nuclear E-cadherin overexpression	10/51 (19.6%)	32/49 (65.3%)	<0.001
Cytoplasmic E-cadherin expression	13/51 (25.4%)	20/56 (35.71%)	0.252
Vimentin expression	16/51 (31.4%)	38/56 (67.9%)	<0.001
Twist	0/51 (0%)	22/55 (40%)	<0.001
Zeb1	10/51 (19.6%)	26/54 (48.1%)	0.004
Nuclear beta-catenin	0/51 (0%)	16/48 (33.3%)	<0.001
Podoplanin (%), mean (SD)	10.54 (20.1)	37.08 (37)	0.001

SD, standard deviation. Bold values correspond to $p < 0.05$ (statistically significant).

DNA methylation of dermal MSCs in psoriasis: Identification of epigenetically dysregulated genes

Mesenchymal stem cells (MSCs) are likely involved in pathological processes of immunerelated diseases, including psoriasis, because of their immunoregulatory and pro-angiogenic effects. DNA methylation plays an essential role in regulating gene expression and maintaining cell function. Hou et al investigated the gene methylation profile of dermal MSCs from patients with psoriasis. They isolated and expanded dermal MSCs from psoriatic patients and normal controls using the attachment assay and conducted genome-wide DNA methylation profile and gene ontology analyses using microarray. The cultured cells were indentified as MSCs by surface marker and differentiation assays. The promoter methylation profile of MSCs from psoriatic derma was markedly different from the normal derma derived MSCs. Genes involved in cell communication, surface receptor signaling pathway, cellular response to stimulus, and cell migration were differently methylated. They found that several aberrantly methylated genes related epidermal proliferation, angiogenesis, and inflammation were differently expressed in psoriatic patients. The MSCs from dermis of psoriasis may participant in the pathogenesis and development of psoriasis through an extraordinarily complex mechanism.

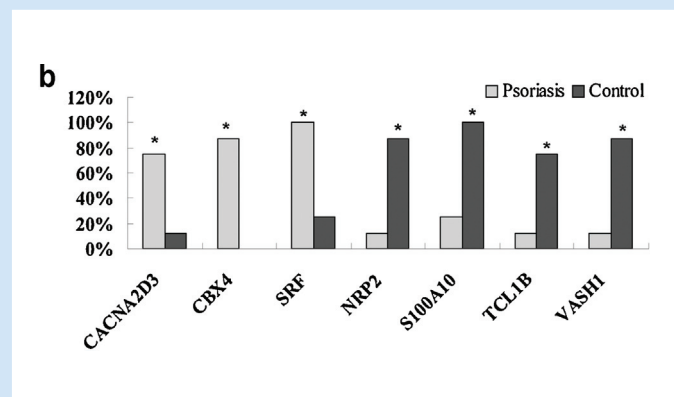


Fig. 3.(b) methylation rate of several genes in the psoriasis and control groups, * $p < 0.05$

Epidermal hyperplasia induced by Raf-MAPK signaling requires Stat3 activation

Raf is one of the downstream effectors of Ras GTPases, and plays a key role in regulating cell proliferation and differentiation through the activation of MAPK. The authors' group previously demonstrated that temporal induction of Raf in the epidermis of K14-Raf:ER transgenic mice results in epidermal hyperplasia resembling squamous cell carcinoma and psoriasis. Meanwhile, epidermal Stat3 activation is required for psoriasis development. Trutani et al investigated whether Stat3 signaling is involved in Raf-MAPK-dependent epidermal hyperplasia. K14-Raf:ER transgenic mice, in which the 4-hydroxytamoxifen (4OHT)-responsive mutant estrogen receptor ligand binding domain-Raf fusion gene is expressed under control of the keratin 14 promoter, were mated with epidermis-specific Stat3 null mice (K5-Cre.Stat3flox/flox). K5-Cre.Stat3flox/flox mice were used to define the impact of Stat3 deficiency on Raf-induced epidermal hyperplasia. They found that over-expression of Raf by 4OHT treatment in K14-Raf:ER;K5-Cre.Stat3flox/flox mice greatly attenuated the epidermal hyperplasia and dermal cell infiltrates compared with K14-Raf:ER;K5-Cre.Stat3flox/WT mice. Also, up-regulation

of psoriasis-associated cytokine profiles, including VEGF, was inhibited in the skin from K14-Raf:ER;K5-Cre.Stat3flox/flox mice following 4OHT treatment. Therefore, Raf-MAPK-dependent psoriatic-like epidermal hyperplasia requires Stat3 signaling in keratinocytes.

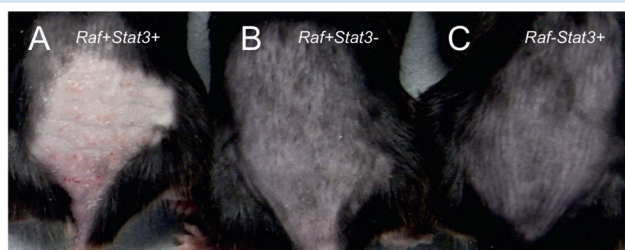


Fig. 3. Stat3 deficient K14:Raf-ER mice did not show epidermal hyperplasia following treatment with 4OHT. Gross appearance of Raf+Stat3+ (A), Raf+Stat3- (B) and Raf-Stat3+ (C, wild-type control) mice at 7 d after topical treatment with 4OHT. D, E, F show histological views of the back skins from A, B, C, respectively. Bar: 20 μ m.

Regulation of lipid production by acetylcholine signalling in human sebaceous glands

The extraneuronal cholinergic system may have numerous functions in the skin, such as terminal differentiation, barrier formation, sweat secretion and the microcirculation.

However, the evidence for cholinergic signalling in sebaceous glands is lacking.

Li et al investigated the role of acetylcholine signalling in sebaceous glands using human sebocytes and a clinical study using botulinum toxin. They demonstrated that human skin sebaceous glands *in vivo* and sebocytes *in vitro* express nicotinic acetylcholine receptor $\alpha 7$ (nAChR $\alpha 7$), and that acetylcholine increased lipid synthesis. When sebocytes were incubated with α -bungarotoxin, a competitive nAChR antagonist, acetylcholine failed to up-regulate lipid synthesis. Twenty healthy volunteers were enrolled in a double-blind, placebo-controlled, split-face study. A marked decrease in sebum production on the botulinum-treated side was found in volunteers with oily skin. Thus, acetylcholine signalling plays a significant role in human sebaceous gland biology and identify

acetylcholine signalling as a promising target in the clinical management of disorders in which sebum production is increased, such as acne vulgaris.

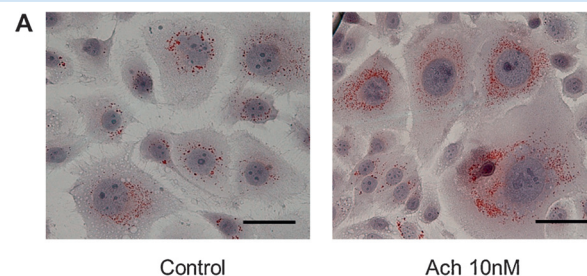


Fig. 2. Increased lipid synthesis in sebocytes in response to Ach. (A) Detection of intracellular lipids in sebocytes treated with 10 nM Ach by microscopy after Oil Red O staining. Scale bars = 20 μ m.

Induction of chemokine receptor CXCR4 expression by transforming growth factor- β 1 in human basal cell carcinoma cells

Higher CXCR4 expression enhances basal cell carcinoma (BCC) invasion and angiogenesis. Chu et al investigated the mechanisms involved in the regulation of CXCR4 expression in invasive BCC. They found that invasive BCC has higher TGF- β 1 and CTGF levels compared to non-invasive BCC. Non-contact dermal fibroblast co-culture with human BCC cells also increased the expression of CXCR4 in BCC cells. Treatment with recombinant human TGF- β 1, but not CTGF, enhanced the CXCR4 levels. The protein level and surface expression of CXCR4 in human BCC cells was increased by TGF- β 1 treatment. TGF- β 1 was intensely expressed in the surrounding fibroblasts of invasive BCC and was positively correlated with the CXCR4 expression of BCC cells. The transcriptional regulation of CXCR4 by TGF- β 1 is mediated by its binding to the TGF- β receptor II and phosphorylation of the extracellular signal-related kinase 1/2 (ERK1/2)-ETS-1 pathway. Thus, TGF- β 1 induces upregulation of CXCR4 in human BCC cells by phosphorylation of ERK1/2-ETS-1 pathway.

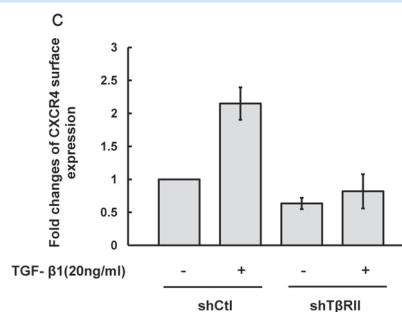


Fig. 4. TGF- β 1-induced upregulation of CXCR4 in human BCC cells is dependent upon TGF- β 1-T β RII interaction and the ERK signaling pathway. Flow cytometric analysis showed that TGF- β 1-induced surface expression of CXCR4 in human BCC cells was abolished by shT β RII, but was unaffected by shCtl. Data are presented as relative changes to the control and are shown as mean $\pm$ S.D. of three independent experiments. * P < 0.05, Student's *t*-test.

Detection of hair-microRNAs as the novel potent biomarker: Evaluation of the usefulness for the diagnosis of scleroderma

Serum microRNA levels are known as useful biomarkers for various diseases. Recent publication has indicated the existence of microRNAs in hair roots and hair shafts. Wang et al evaluated several methods for the extraction of hair microRNAs, and their usefulness for the diagnosis of scleroderma. They found that microRNAs in hair roots and hair shafts were detectable and quantitative using their method.

The difference of microRNA levels in hair roots and hair shafts obtained from different places of head in each individual were within 2-fold, indicating the reproducibility of hair microRNA levels by the method. PCR array revealed microRNAs from sera, hair roots and hair shafts have different expression pattern, and can be independent biomarkers. Serum and hair root miR-196a levels were not significantly changed in scleroderma patients, while miR-196a levels in hair shafts were significantly decreased in scleroderma patient. Thus, hairs are more accessible than sera among human samples. microRNAs levels in hair roots or hair shafts may become effective and independent biomarkers.